

# Specialist Mathematics · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name \_\_\_\_\_ Date \_\_\_\_\_

Exam 1 ·

**39 marks**

reading 15 min · writing 60 min

Exam 2 ·

**79 marks**

reading 15 min · writing 120 min

State average on these questions

**85 / 118 (72%)**

a student at the state average on every question

## How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g.  $v = \sqrt{2gh} = 9.9 \text{ m/s}$ .
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

## Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the easiest questions.

## Sources

- [2023 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2023 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)

## Exam 1 · technology free (39 marks, reading 15 min, writing 60 min)

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### Questions (technology free)

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Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

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#### Question 1 (4 marks)

2023 VCAA Exam 1 · Question 1 · [open in the official exam PDF \(p. 3\)](#) ↗

Rational function  $f(x) = \frac{x^2+x-6}{x-1}$ : rewrite by division, then sketch with asymptotes

a. Show  $f(x) = x + 2 - \frac{4}{x-1}$  (1 mark)

b. Sketch the graph, labelling asymptotes (3 marks)

#### Question 2 (3 marks)

2023 VCAA Exam 1 · Question 5 · [open in the official exam PDF \(p. 5\)](#) ↗

Evaluate  $\int_1^2 x^2 \log_e x \, dx$

Evaluate the integral (3 marks)

#### Question 3 (4 marks)

2023 VCAA Exam 1 · Question 6 · [open in the official exam PDF \(p. 6\)](#) ↗

Sum of three independent normal travel times; probability statement for a 12-day average waiting time

a. Mean and sd of the total time (2 marks)

b. Values  $a, b$  for the 12-day average (2 marks)

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**Question 4** (6 marks)

2023 VCAA Exam 1 · Question 9 · [open in the official exam PDF \(p. 9\)](#) ↗

Plane through three points with an unknown coordinate: y-intercept, vectors in the plane, Cartesian equation, the unknown, parallelogram area

a. Coordinates of the y-intercept D (1 mark)

b. Show the vectors  $\overrightarrow{AB}$  and  $\overrightarrow{AD}$  (1 mark)

c. Cartesian equation of the plane (2 marks)

d. Find  $a$  (1 mark)

e. Area of the parallelogram (1 mark)

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**Question 5** (3 marks)

2024 VCAA Exam 1 · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Direct proof that  $2x^2 - 3x - 7$  is even whenever  $x$  is odd

Direct proof (3 marks)

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**Question 6** (4 marks)

2024 VCAA Exam 1 · Question 4 · [open in the official exam PDF \(p. 6\)](#) ↗

Angle between two vectors; integer  $n$  for which a dot product equals the magnitude of a cross product

a. Angle between  $\vec{a}$  and  $\vec{b}$  (2 marks)

b. Values of  $n$  (2 marks)

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**Question 7** (3 marks)

2024 VCAA Exam 1 · Question 5 · [open in the official exam PDF \(p. 7\)](#) ↗

Volume of revolution of  $y = \sqrt{k - \frac{1}{x^2}}$  leads to a cubic in  $k$

Show  $k$  satisfies the cubic (3 marks)

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**Question 8** (4 marks)

2025 VCAA Exam 1 · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Tangent to  $xe^{-2y} + y^2e^x = 8e^4$  at  $(4, -2)$

Equation of the tangent (4 marks)

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**Question 9** (3 marks)

2025 VCAA Exam 1 · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Point of intersection of two lines in 3D given by a point and a direction each

Coordinates of the intersection (3 marks)

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**Question 10** (5 marks)

2025 VCAA Exam 1 · Question 8 · [open in the official exam PDF \(p. 9\)](#) ↗

Quartic  $z^4 + 6z^2 + 25$ : plot  $1 + 2i$  and its conjugate, a quadratic factor, the remaining roots

**a.** Plot and label  $z_1$  and  $\bar{z}_1$  (1 mark)

**b.** A quadratic factor (2 marks)

**c.** Remaining solutions (2 marks)

## Exam 2 · CAS and a bound reference allowed (79 marks, reading 15 min, writing 120 min)

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### Section A – Multiple-choice questions

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Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

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#### Question 1 · 1 mark

2023 VCAA Exam 2 · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Contrapositive of an “if ... then ...” statement about a football team

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

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#### Question 2 · 1 mark

2023 VCAA Exam 2 · Section A · Question 3 · [open in the official exam PDF \(p. 2\)](#) ↗

Values of  $a$  for which  $y = a + \sec x$  on  $[-\pi, \pi]$  has two  $x$ -intercepts

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

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#### Question 3 · 1 mark

2023 VCAA Exam 2 · Section A · Question 6 · [open in the official exam PDF \(p. 3\)](#) ↗

Pseudocode implementing Euler's method: after how many iterations a given value is printed

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

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#### Question 4 · 1 mark

2023 VCAA Exam 2 · Section A · Question 7 · [open in the official exam PDF \(p. 4\)](#) ↗

Reading a direction field to estimate  $y$  on a solution curve

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

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#### Question 5 · 1 mark

2023 VCAA Exam 2 · Section A · Question 17 · [open in the official exam PDF \(p. 8\)](#) ↗

Find  $\alpha$ ,  $\beta$ ,  $\gamma$  from  $\underset{\sim}{a} \times \underset{\sim}{b} = \underset{\sim}{c}$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

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#### Question 6 · 1 mark

2023 VCAA Exam 2 · Section A · Question 18 · [open in the official exam PDF \(p. 8\)](#) ↗

Value of  $k$  making two planes perpendicular

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

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#### Question 7 · 1 mark

2024 VCAA Exam 2 · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Contrapositive of a statement about integers  $m$ ,  $n$  with  $m + n \geq 9$

Your answer: ☐ A ☐ B ☐ C ☐ D

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#### Question 8 · 1 mark

2024 VCAA Exam 2 · Section A · Question 3 · [open in the official exam PDF \(p. 2\)](#) ↗

Values of  $h$  for which  $\frac{x-h}{(x+1)(x-4)}$  has no turning points

Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 9** · 1 mark2024 VCAA Exam 2 · Section A · Question 5 · [open in the official exam PDF \(p. 3\)](#) ↗Geometric transformation taking  $z$  to  $-\bar{z}$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 10** · 1 mark2024 VCAA Exam 2 · Section A · Question 6 · [open in the official exam PDF \(p. 3\)](#) ↗Value of  $k$  making  $z^2 + 4iz + 3$  purely imaginary for  $z = 3 + ki$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 11** · 1 mark2024 VCAA Exam 2 · Section A · Question 7 · [open in the official exam PDF \(p. 4\)](#) ↗Separating variables in  $\frac{dy}{dx} = e^{x-y}(\cos(x-y) - \cos(x+y))$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 12** · 1 mark2024 VCAA Exam 2 · Section A · Question 10 · [open in the official exam PDF \(p. 5\)](#) ↗Surface area when a parametric line segment is rotated about the  $y$ -axisYour answer: ☐ A ☐ B ☐ C ☐ D

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**Question 13** · 1 mark2024 VCAA Exam 2 · Section A · Question 13 · [open in the official exam PDF \(p. 6\)](#) ↗

Unknown component from the angle between two vectors

Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 14** · 1 mark2025 VCAA Exam 2 · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Contrapositive of "if I have a tiger, then I have a cat"

Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 15** · 1 mark2025 VCAA Exam 2 · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Output of an algorithm summing disc volumes

Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 16** · 1 mark2025 VCAA Exam 2 · Section A · Question 5 · [open in the official exam PDF \(p. 4\)](#) ↗Product  $ab$  for a real cubic with root  $2 - 3i$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 17** · 1 mark2025 VCAA Exam 2 · Section A · Question 6 · [open in the official exam PDF \(p. 4\)](#) ↗ $\operatorname{Re}\left(\frac{1}{1-z}\right)$  when  $|z| = 1$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 18** · 1 mark2025 VCAA Exam 2 · Section A · Question 7 · [open in the official exam PDF \(p. 4\)](#) ↗Rewriting a trig integral with the substitution  $u = \cos \theta$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 19** · 1 mark2025 VCAA Exam 2 · Section A · Question 8 · [open in the official exam PDF \(p. 5\)](#) ↗

Matching a direction field to its differential equation

Your answer: ☐ A ☐ B ☐ C ☐ D

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**Question 20** · 1 mark2025 VCAA Exam 2 · Section A · Question 10 · [open in the official exam PDF \(p. 6\)](#) ↗Upper limit  $a$  for a volume about the  $y$ -axis involving  $y = 3 \cos^{-1} x$ Your answer: ☐ A ☐ B ☐ C ☐ D

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**Section B – Extended-response questions**

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Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

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**Question 1** (10 marks)2023 VCAA Exam 2 · Section B · Question 1 · [open in the official exam PDF \(p. 10\)](#) ↗

Walking track made of a hybrid function, then an elliptical return path: parameters, smoothness, turning and inflection points, Cartesian ellipse, path length

**a.** Show  $a = -1$  and  $b = 0$  (1 mark)**b.** Verify the curves meet smoothly at C (2 marks)**c.i.** Coordinates of A (1 mark)**c.ii.** Coordinates of B (1 mark)

**d.** Cartesian equation of the ellipse (2 marks)

**e.** Sketch the elliptical path from D to O (1 mark)

**f.i.** Integral for the length of the path (1 mark)

**f.ii.** Length to 3 d.p. (1 mark)

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**Question 2** (9 marks)

2023 VCAA Exam 2 · Section B · Question 6 · [open in the official exam PDF \(p. 22\)](#) ↗

Koala masses: a 95% confidence interval, expected capture count, sample size to shrink the interval, a one-tailed test with p-value, critical value and type II error

**a.** 95% confidence interval (1 mark)

**b.** How many of 60 intervals contain  $\mu$  (1 mark)

**c.** Sample size for a 60% narrower interval (1 mark)

**d.** Hypotheses (1 mark)

**e.i.** p-value to 4 d.p. (1 mark)

**e.ii.** Conclusion with a reason (1 mark)

**f.** Critical sample mean to 3 d.p. (1 mark)

**g.** Probability of a type II error (1 mark)

**h.** Label the critical value and shade the type II region (1 mark)

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**Question 3** (10 marks)

2024 VCAA Exam 2 · Section B · Question 2 · [open in the official exam PDF \(p. 12\)](#) ↗

Loci in the complex plane: a perpendicular bisector, a circle on a diameter, sketching a circle, a ray, and a minor segment area

**a.**  $|z - z_1| = |z - z_2|$  as  $y = mx + c$  (2 marks)

**b.** Equation of the circle (2 marks)

**c.** Sketch the circle (2 marks)

**d.i.** Sketch the ray (1 mark)

**d.ii.** Equation of the ray (1 mark)

**e.** Area of the minor segment (2 marks)

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**Question 4** (10 marks)

2025 VCAA Exam 2 · Section B · Question 1 · [open in the official exam PDF \(p. 12\)](#) ↗

$y = \frac{3x}{x^3+x+2}$ : sketch with turning point and inflection, a volume, asymptotes of a related curve, then the family  $\frac{3x}{x^3+ax+2}$

**a.** Sketch with asymptotes, turning point and inflection (3 marks)

**b.i.** Integral for the volume (1 mark)

**b.ii.** Volume to 2 d.p. (1 mark)

**c.** Vertical asymptotes of  $\frac{3x}{x^3-5x+2}$  (1 mark)

**d.i.** y-coordinate of the stationary point P (1 mark)

**d.ii.** Vertical asymptotes when there are no stationary points (1 mark)

**d.iii.** a for an inflection at  $x = 2$  (2 marks)

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**Question 5** (10 marks)

2025 VCAA Exam 2 · Section B · Question 2 · [open in the official exam PDF \(p. 14\)](#) ↗

Loci on the Argand plane: a circle, a perpendicular bisector, their intersections, a ray and a minor segment

**a.** Sketch  $z\bar{z} = 4$  (1 mark)

**b.i.** Show the locus is  $y = \sqrt{3}x$  (2 marks)

**b.ii.** Sketch the line (1 mark)

**c.i.** Intersection points (2 marks)

**c.ii.** Label the points (1 mark)

**d.**  $z_0$  and  $\theta$  for the ray from P through Q (1 mark)

**e.** Area of the minor segment (2 marks)

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**Question 6** (10 marks)

2025 VCAA Exam 2 · Section B · Question 6 · [open in the official exam PDF \(p. 22\)](#) ↗

Water consumption and bottle filling: sampling distribution, a confidence interval, sample size, then a one-tailed test with critical value and type II error

**a.i.** Mean and sd of the sample mean (1 mark)

**a.ii.** Probability the mean exceeds 970 (1 mark)

**b.** 95% confidence interval (1 mark)

**c.** How many of 300 intervals contain  $\mu$  (1 mark)

**d.** Minimum sample size (1 mark)

**e.** Hypotheses (1 mark)

**f.i.** p-value (1 mark)

**f.ii.** Is the company's claim correct? (1 mark)

**g.** Critical sample mean (1 mark)

**h.** Probability of a type II error (1 mark)

END OF PAPER

# Marking guide follows

Stop here until you have finished the paper under exam conditions.

## Marking guide · Exam 1

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### Questions (technology free)

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#### Question 1 (4 marks) · state average 2.8 / 4 (70%)

2023 VCAA Exam 1 · Question 1

a. Show  $f(x) = x + 2 - \frac{4}{x-1}$

☐ Division:  $x^2 + x - 6 = (x - 1)(x + 2) - 4$

**Answer / notes:**  $x^2 + x - 6 = (x - 1)(x + 2) - 4$ , so divide through by  $x - 1$ .

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

**Your mark:** \_\_\_\_\_ / 1

b. Sketch the graph, labelling asymptotes

☐ Asymptotes  $x = 1$  and  $y = x + 2$  drawn and labelled

☐ Branches approach both asymptotes (correct shape in each region)

☐ Passes through  $(-3, 0)$ ,  $(2, 0)$  and  $(0, 6)$

**Answer / notes:** Asymptotes  $x = 1$ ,  $y = x + 2$ ; intercepts  $(-3, 0)$ ,  $(2, 0)$ ,  $(0, 6)$ . Examiners saw graphs drifting away from the asymptotes and missing oblique asymptotes.

State average 1.9 / 3 (63%); 39% scored full marks. Mark distribution: 0: 18% · 1: 20% · 2: 23% · 3: 39%

**Your mark:** \_\_\_\_\_ / 3

#### Question 2 (3 marks) · state average 2.0 / 3 (67%)

2023 VCAA Exam 1 · Question 5

Evaluate the integral

☐ Parts with  $u = \log_e x$ ,  $v' = x^2$ :  $\left[ \frac{x^3}{3} \log_e x \right]_1^2 - \int_1^2 \frac{x^2}{3} dx$

☐  $\int_1^2 \frac{x^2}{3} dx = \frac{7}{9}$

☐  $\frac{8}{3} \log_e 2 - \frac{7}{9}$

**Answer / notes:**  $\frac{8}{3} \log_e 2 - \frac{7}{9}$ . Some students chose  $u$  and  $v'$  the wrong way round.

State average 2.0 / 3 (67%); 53% scored full marks. Mark distribution: 0: 23% · 1: 10% · 2: 14% · 3: 53%

**Your mark:** \_\_\_\_\_ / 3

#### Question 3 (4 marks) · state average 2.6 / 4 (65%)

2023 VCAA Exam 1 · Question 6

a. Mean and sd of the total time

☐ Mean  $20 + 8 + 12 = 40$

☐  $sd = \sqrt{36 + 3 + 25} = 8$

**Answer / notes:** Mean 40 min, sd 8 min. Add variances, not standard deviations.

State average 1.7 / 2 (85%); 75% scored full marks. Mark distribution: 0: 6% · 1: 19% · 2: 75%

**Your mark:** \_\_\_\_\_ / 2

b. Values  $a$ ,  $b$  for the 12-day average

☐  $sd(\bar{X}_w) = \frac{\sqrt{3}}{\sqrt{12}} = \frac{1}{2}$

☐  $a = -\frac{1}{2}$ ,  $b = 1$

**Answer / notes:** 7.75 and 8.5 min standardise to  $a = -0.5$ ,  $b = 1$  (by symmetry  $a = -1$ ,  $b = 0.5$  also works). The common error was using  $\sqrt{3}$  instead of  $\frac{\sqrt{3}}{\sqrt{12}}$ .

State average 0.9 / 2 (45%); 40% scored full marks. Mark distribution: 0: 47% · 1: 13% · 2: 40%

**Your mark:** \_\_\_\_\_ / 2

#### Question 4 (6 marks) · state average 4.4 / 6 (73%)

2023 VCAA Exam 1 · Question 9

a. Coordinates of the y-intercept D

☐  $(0, 2, 0)$

**Answer / notes:**  $D(0, 2, 0)$

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

**Your mark:** \_\_\_\_\_ / 1

b. Show the vectors  $\overrightarrow{AB}$  and  $\overrightarrow{AD}$

☐  $\overrightarrow{AB} = \underset{\sim}{b} - \underset{\sim}{a}, \overrightarrow{AD} = \underset{\sim}{d} - \underset{\sim}{a}$  shown

**Answer / notes:** End minus start for each.

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

**Your mark:** \_\_\_\_\_ / 1

c. Cartesian equation of the plane

☐ Normal  $\overrightarrow{AB} \times \overrightarrow{AD} = -4\underset{\sim}{i} - 2\underset{\sim}{j} - 3\underset{\sim}{k}$

☐  $4x + 2y + 3z = 4$

**Answer / notes:**  $4x + 2y + 3z = 4$

State average 1.4 / 2 (70%); 56% scored full marks. Mark distribution: 0: 21% · 1: 23% · 2: 56%

**Your mark:** \_\_\_\_\_ / 2

d. Find a

☐  $a = -\frac{9}{4}$

**Answer / notes:** Substitute C:  $4a - 2 + 15 = 4$ , so  $a = -\frac{9}{4}$ .

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

**Your mark:** \_\_\_\_\_ / 1

e. Area of the parallelogram

☐  $\sqrt{29}$

**Answer / notes:**  $|\overrightarrow{AB} \times \overrightarrow{AD}| = \sqrt{16 + 4 + 9} = \sqrt{29}$

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

**Your mark:** \_\_\_\_\_ / 1

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**Question 5** (3 marks) · state average 2.5 / 3 (83%)

2024 VCAA Exam 1 · Question 2

Direct proof

☐ Lets  $x = 2k + 1, k \in \mathbb{Z}$

☐ Expands to  $8k^2 + 2k - 8$

☐  $= 2(4k^2 + k - 4)$ , a multiple of 2, so even

**Answer / notes:**  $2(2k + 1)^2 - 3(2k + 1) - 7 = 2(4k^2 + k - 4)$ , which is even.

State average 2.5 / 3 (83%); 65% scored full marks. Mark distribution: 0: 8% · 1: 4% · 2: 23% · 3: 65%

**Your mark:** \_\_\_\_\_ / 3

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**Question 6** (4 marks) · state average 2.9 / 4 (72%)

2024 VCAA Exam 1 · Question 4

a. Angle between  $\underset{\sim}{a}$  and  $\underset{\sim}{b}$

☐  $\cos \theta = \frac{-9}{3\sqrt{2} \times 3} = -\frac{1}{\sqrt{2}}$

☐  $\theta = \frac{3\pi}{4}$

**Answer / notes:**  $\frac{3\pi}{4}$  (135°). Many found  $\cos \theta$  correctly but then gave  $\frac{\pi}{4}$ .

State average 1.4 / 2 (70%); 45% scored full marks. Mark distribution: 0: 10% · 1: 45% · 2: 45%

**Your mark:** \_\_\_\_\_ / 2

**b. Values of  $n$**

☐  $\tilde{a} \cdot \tilde{c} = 9$  and  $|\tilde{a} \times \tilde{c}| = \sqrt{9 + 18n^2}$

☐  $n = \pm 2$

**Answer / notes:**  $81 = 9 + 18n^2$ , so  $n = \pm 2$ . Give both values.

State average 1.5 / 2 (75%); 60% scored full marks. Mark distribution: 0: 11% · 1: 29% · 2: 60%

**Your mark:** \_\_\_\_\_ / 2

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**Question 7** (3 marks) · state average 2.1 / 3 (70%)

2024 VCAA Exam 1 · Question 5

Show  $k$  satisfies the cubic

☐  $V = \pi \int_1^{k/2} \left(k - \frac{1}{x^2}\right) dx$

☐  $= \pi \left(\frac{k^2}{2} + \frac{2}{k} - k - 1\right)$

☐ Sets equal to  $\frac{7\pi}{2}$  and rearranges to  $k^3 - 2k^2 - 9k + 4 = 0$

**Answer / notes:**  $\pi \left[kx + \frac{1}{x}\right]_1^{k/2} = \frac{7\pi}{2}$ , then multiply by  $2k$ . Some students used a surface-area formula by mistake.

State average 2.1 / 3 (70%); 59% scored full marks. Mark distribution: 0: 22% · 1: 11% · 2: 9% · 3: 59%

**Your mark:** \_\_\_\_\_ / 3

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**Question 8** (4 marks) · state average 2.9 / 4 (72%)

2025 VCAA Exam 1 · Question 1

Equation of the tangent

☐ Implicit differentiation:  $e^{-2y} - 2xe^{-2y}\frac{dy}{dx} + 2ye^x\frac{dy}{dx} + y^2e^x = 0$

☐ Substitutes  $(4, -2)$ :  $5e^4 - 12e^4\frac{dy}{dx} = 0$

☐  $\frac{dy}{dx} = \frac{5}{12}$

☐  $y = \frac{5}{12}x - \frac{11}{3}$

**Answer / notes:**  $y + 2 = \frac{5}{12}(x - 4)$ , i.e.  $y = \frac{5x}{12} - \frac{11}{3}$ . Arithmetic slips in the gradient cost most marks.

State average 2.9 / 4 (72%); 48% scored full marks. Mark distribution: 0: 10% · 1: 3% · 2: 27% · 3: 12% · 4: 48%

**Your mark:** \_\_\_\_\_ / 4

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**Question 9** (3 marks) · state average 2.2 / 3 (73%)

2025 VCAA Exam 1 · Question 2

Coordinates of the intersection

☐ Parametric forms with different parameters, e.g.  $(2 + s, 3 + 2s, 1 - s) = (1 - t, 3 - t, 2 + t)$

☐ Solves two equations:  $s = 1$ ,  $t = -2$  (checks the third)

☐  $(3, 5, 0)$

**Answer / notes:**  $(3, 5, 0)$ . Use a different parameter for each line.

State average 2.2 / 3 (73%); 59% scored full marks. Mark distribution: 0: 14% · 1: 13% · 2: 15% · 3: 59%

**Your mark:** \_\_\_\_\_ / 3

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**Question 10** (5 marks) · state average 3.3 / 5 (66%)

2025 VCAA Exam 1 · Question 8

**a. Plot and label  $z_1$  and  $\bar{z}_1$**

☐  $(1, 2)$  and  $(1, -2)$  plotted and labelled

**Answer / notes:**  $1 + 2i$  and  $1 - 2i$ .

State average 0.8 / 1 (80%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

**Your mark:** \_\_\_\_\_ / 1

**b.** A quadratic factor

☐ Conjugate  $1 - 2i$  is also a root

☐  $z^2 - 2z + 5$

**Answer / notes:**  $(z - 1 - 2i)(z - 1 + 2i) = z^2 - 2z + 5$ .

State average 1.4 / 2 (70%); 61% scored full marks. Mark distribution: 0: 19% · 1: 20% · 2: 61%

**Your mark:** \_\_\_\_\_ / 2

**c.** Remaining solutions

☐ Other factor  $z^2 + 2z + 5$

☐  $z = -1 \pm 2i$

**Answer / notes:**  $z^4 + 6z^2 + 25 = (z^2 - 2z + 5)(z^2 + 2z + 5)$ , so  $z = -1 \pm 2i$ .

State average 1.1 / 2 (55%); 42% scored full marks. Mark distribution: 0: 37% · 1: 21% · 2: 42%

**Your mark:** \_\_\_\_\_ / 2

## Marking guide · Exam 2

### Section A – Multiple-choice questions

| ✓                        | Q  | Ans | Why                                                                                                                                                                                                                                                                                       | How Victoria went                                  |
|--------------------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <input type="checkbox"/> | 1  | C   | Contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$ : “if they are training enough, then the team does not play badly”.                                                                                                                                                    | 85% correct<br>A 2% · B 1% · C 85% · D 6% · E 5%   |
| <input type="checkbox"/> | 2  | E   | Solve $\sec x = -a$ . Two solutions when $-a > 1$ or $-a \leq -1$ (at $\sec x = -1$ both endpoints $\pm\pi$ count). So $a < -1$ or $a \geq 1$ .                                                                                                                                           | 72% correct<br>A 3% · B 6% · C 14% · D 4% · E 72%  |
| <input type="checkbox"/> | 3  | C   | Step by hand: $y = 0.5$ , then $0.5 + 0.5e^{0.25} \approx 1.142$ , then $1.142 + 0.5e^{1.142} \approx 2.709$ : the third iteration.                                                                                                                                                       | 69% correct<br>A 2% · B 12% · C 69% · D 14% · E 2% |
| <input type="checkbox"/> | 4  | D   | Sketch the curve through the given point by following the segments, then read off its height at the required $x$ .                                                                                                                                                                        | 68% correct<br>A 5% · B 3% · C 14% · D 68% · E 10% |
| <input type="checkbox"/> | 5  | C   | $\underset{\sim}{a} \times \underset{\sim}{b} = (4 + \underset{\sim}{\beta})\underset{\sim}{i} - (4\underset{\sim}{\alpha} + 3)\underset{\sim}{j} + (\underset{\sim}{\alpha}\underset{\sim}{\beta} - 3)\underset{\sim}{k}$ . Matching gives $\beta = -2$ , $\alpha = 1$ , $\gamma = -5$ . | 73% correct<br>A 5% · B 8% · C 73% · D 6% · E 7%   |
| <input type="checkbox"/> | 6  | C   | Normals must be perpendicular: $(2, -k, 3) \cdot (2k, 3, -2) = k - 6 = 0$ .                                                                                                                                                                                                               | 74% correct<br>A 6% · B 10% · C 74% · D 7% · E 3%  |
| <input type="checkbox"/> | 7  | C   | Negate both parts and swap: “if $m < 5$ and $n < 5$ , then $m + n < 9$ ”. Negating “or” gives “and”: the popular wrong answer kept “or”. <i>Most popular wrong answer: A (20%)</i> .                                                                                                      | 72% correct<br>A 20% · B 3% · C 72% · D 6%         |
| <input type="checkbox"/> | 8  | C   | The numerator of $f'$ is $-x^2 + 2hx - 4 - 3h$ ; no real roots needs $h^2 - 3h - 4 < 0$ . At $h = -1$ or $4$ the function cancels to a reciprocal with no turning points, so $-1 \leq h \leq 4$ .                                                                                         | 70% correct<br>A 13% · B 11% · C 70% · D 6%        |
| <input type="checkbox"/> | 9  | C   | $-\bar{z}$ keeps the imaginary part and negates the real part: a reflection in the imaginary axis.                                                                                                                                                                                        | 70% correct<br>A 13% · B 13% · C 70% · D 4%        |
| <input type="checkbox"/> | 10 | D   | Real part $12 - 4k - k^2 = 0$ gives $k = 2$ or $-6$ ; only $k = 2$ is offered (and the imaginary part is non-zero).                                                                                                                                                                       | 77% correct<br>A 12% · B 6% · C 5% · D 77%         |
| <input type="checkbox"/> | 11 | C   | $\cos(x - y) - \cos(x + y) = 2 \sin x \sin y$ , so $\int \frac{e^y}{\sin y} dy = 2 \int e^x \sin x dx$ . <i>Most popular wrong answer: B (15%)</i> .                                                                                                                                      | 72% correct<br>A 7% · B 15% · C 72% · D 6%         |
| <input type="checkbox"/> | 12 | A   | $S = 2\pi \int_0^k x \sqrt{\dot{x}^2 + \dot{y}^2} dt = 2\pi \int_0^k 5t \cdot 13 dt = 65\pi k^2$ . <i>Most popular wrong answer: C (15%)</i> .                                                                                                                                            | 74% correct<br>A 74% · B 9% · C 15% · D 2%         |
| <input type="checkbox"/> | 13 | B   | $\frac{16-m}{3\sqrt{40+m^2}} = \frac{13}{21}$ . Testing $m = 3$ : $\frac{13}{3 \times 7} \checkmark$ .                                                                                                                                                                                    | 81% correct<br>A 4% · B 81% · C 10% · D 4%         |
| <input type="checkbox"/> | 14 | C   | $\neg Q \Rightarrow \neg P$ : “if I do not have a cat, then I do not have a tiger”.                                                                                                                                                                                                       | 93% correct<br>A 5% · B 2% · C 93% · D 0%          |
| <input type="checkbox"/> | 15 | B   | $\pi(f(1)^2 + f(2)^2 + f(3)^2) = \pi(2 + 3 + 4) = 9\pi$ . <i>Most popular wrong answer: C (15%)</i> .                                                                                                                                                                                     | 72% correct<br>A 9% · B 72% · C 15% · D 3%         |

| ✓                        | Q  | Ans | Why                                                                                                                                                                                                         | How Victoria went                           |
|--------------------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <input type="checkbox"/> | 16 | A   | Conjugate pair gives $z^2 - 4z + 13$ ; constant term $-52$ means the third factor is $z - 4$ . Expanding: $a = -8$ , $b = 29$ , $ab = -232$ . <i>Most popular wrong answer: B (16%)</i> .                   | 69% correct<br>A 69% · B 16% · C 9% · D 6%  |
| <input type="checkbox"/> | 17 | C   | With $z = \operatorname{cis} \theta$ , $\frac{1}{1-z} = \frac{1-\bar{z}}{ 1-z ^2}$ and $ 1-z ^2 = 2 - 2\cos \theta$ , giving real part $\frac{1}{2}$ .                                                      | 70% correct<br>A 8% · B 11% · C 70% · D 10% |
| <input type="checkbox"/> | 18 | D   | $\frac{1}{2} \sin 2\theta = \sin \theta \cos \theta$ ; with $du = -\sin \theta d\theta$ the terminals flip to 0 $\rightarrow$ 1: $\int_0^1 \frac{u}{1+u} du = \int_0^1 \left(1 - \frac{1}{1+u}\right) du$ . | 79% correct<br>A 4% · B 11% · C 6% · D 79%  |
| <input type="checkbox"/> | 19 | A   | Gradients are symmetric in x (an $x^2$ term) and negative on the positive y-axis: $\frac{dy}{dx} = x^2 - y$ .                                                                                               | 80% correct<br>A 80% · B 10% · C 4% · D 6%  |
| <input type="checkbox"/> | 20 | D   | $x = \cos \frac{y}{3}$ : $V = \pi \int_0^a \cos^2 \frac{y}{3} dy = \frac{\pi}{2} \left(a + \frac{3}{2} \sin \frac{2a}{3}\right)$ . $a = \pi$ matches.                                                       | 69% correct<br>A 8% · B 12% · C 11% · D 69% |

Multiple-choice score:  / 20

## Section B – Extended-response questions

**Question 1** (10 marks) · state average 7.3 / 10 (73%)

2023 VCAA Exam 2 · Section B · Question 1

a. Show  $a = -1$  and  $b = 0$

☐ Uses  $f(1) = 0$  for both pieces

**Answer / notes:**  $-(1+a)^2 = 0$  and  $e^0 - 1 + b = 0$ .

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

**Your mark:**  / 1

b. Verify the curves meet smoothly at C

☐ Both pieces equal 0 at  $x = 1$

☐ Both derivatives equal 0 at  $x = 1$

**Answer / notes:** Show the values match *and* the gradients match. Many only showed they met.

State average 1.4 / 2 (70%); 61% scored full marks. Mark distribution: 0: 27% · 1: 13% · 2: 61%

**Your mark:**  / 2

c.i. Coordinates of A

☐  $\left(\frac{1}{3}, -\frac{4}{27}\right)$

**Answer / notes:**  $f'(x) = -(x-1)(3x-1) = 0$  gives  $x = \frac{1}{3}$ .

State average 0.9 / 1 (90%); 92% scored full marks. Mark distribution: 0: 8% · 1: 92%

**Your mark:**  / 1

c.ii. Coordinates of B

☐  $\left(\frac{2}{3}, -\frac{2}{27}\right)$

**Answer / notes:**  $f''(x) = -(6x-4) = 0$  gives  $x = \frac{2}{3}$ .

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

**Your mark:**  / 1

d. Cartesian equation of the ellipse

☐ Isolates  $\cos t = \frac{x-2}{2}$ ,  $\sin t = \frac{y}{e-2}$

☐  $\frac{(x-2)^2}{4} + \frac{y^2}{(e-2)^2} = 1$

**Answer / notes:**  $\frac{(x-2)^2}{4} + \frac{y^2}{(e-2)^2} = 1$ .

State average 1.7 / 2 (85%); 81% scored full marks. Mark distribution: 0: 11% · 1: 8% · 2: 81%

**Your mark:** \_\_\_\_\_ / 2

e. Sketch the elliptical path from D to O

☐ Quarter ellipse from  $D(2, e-2)$  to  $O$ , horizontal at D and vertical at O

**Answer / notes:** Only 20% of students got this: the curve must be horizontal at D and vertical at O.

State average 0.2 / 1 (20%); 22% scored full marks. Mark distribution: 0: 78% · 1: 22%

**Your mark:** \_\_\_\_\_ / 1

f.i. Integral for the length of the path

☐  $\int_{\pi/2}^{\pi} \sqrt{4 \sin^2 t + (e-2)^2 \cos^2 t} dt$

**Answer / notes:** Terminals  $\frac{\pi}{2}$  to  $\pi$ , not 0 to 2.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

**Your mark:** \_\_\_\_\_ / 1

f.ii. Length to 3 d.p.

☐ 2.255 km

**Answer / notes:**  $\approx 2.255$  km.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

**Your mark:** \_\_\_\_\_ / 1

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**Question 2** (9 marks) · state average 6.2 / 9 (69%)

2023 VCAA Exam 2 · Section B · Question 6

a. 95% confidence interval

☐ (10.95, 11.83)

**Answer / notes:**  $11.39 \pm 1.96 \times \frac{1}{\sqrt{20}}$ .

State average 0.8 / 1 (80%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

**Your mark:** \_\_\_\_\_ / 1

b. How many of 60 intervals contain  $\mu$

☐ 57

**Answer / notes:**  $0.95 \times 60 = 57$ .

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

**Your mark:** \_\_\_\_\_ / 1

c. Sample size for a 60% narrower interval

☐ 125

**Answer / notes:** Width  $\times 0.4$  needs  $n \div 0.4^2$ :  $20 \div 0.16 = 125$ . Only 30% got this.

State average 0.3 / 1 (30%); 28% scored full marks. Mark distribution: 0: 72% · 1: 28%

**Your mark:** \_\_\_\_\_ / 1

d. Hypotheses

☐  $H_0 : \mu = 12$ ,  $H_1 : \mu < 12$

**Answer / notes:** Use  $\mu$ , not  $\bar{x}$ .

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

**Your mark:** \_\_\_\_\_ / 1

e.i. p-value to 4 d.p.

☐ 0.0057

**Answer / notes:**  $\Pr(\bar{X} < 11.6)$  with  $\bar{X} \sim N(12, \frac{1}{40})$ .

State average 0.8 / 1 (80%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

**Your mark:** \_\_\_\_\_ / 1

e.ii. Conclusion with a reason

☐ Reject  $H_0$  since  $p < 0.01$

**Answer / notes:** Refer to the p-value in your reason.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

**Your mark:** \_\_\_\_\_ / 1

f. Critical sample mean to 3 d.p.

☐ 11.632

**Answer / notes:**  $12 - 2.3263 \times \frac{1}{\sqrt{40}} \approx 11.632$ .

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

**Your mark:** \_\_\_\_\_ / 1

g. Probability of a type II error

☐ 0.071

**Answer / notes:**  $\Pr(\bar{X} > 11.632 \mid \mu = 11.4) \approx 0.071$ .

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

**Your mark:** \_\_\_\_\_ / 1

h. Label the critical value and shade the type II region

☐ (Question invalidated by VCAA; all students awarded the mark)

**Answer / notes:** VCAA invalidated this part because of an error in the diagram. The type II region is the area under the  $H_1$  curve to the right of the critical value.

State average 1.0 / 1 (100%); 100% scored full marks. Mark distribution: 0: 0% · 1: 100%

**Your mark:** \_\_\_\_\_ / 1

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**Question 3** (10 marks) · state average 6.9 / 10 (69%)

2024 VCAA Exam 2 · Section B · Question 2

a.  $|z - z_1| = |z - z_2|$  as  $y = mx + c$

☐ Expands with  $z = x + iy$ :  $6x - 4y = 11$

☐  $y = \frac{3}{2}x - \frac{11}{4}$

**Answer / notes:** This is the perpendicular bisector of  $z_1z_2$ .

State average 1.5 / 2 (75%); 63% scored full marks. Mark distribution: 0: 13% · 1: 24% · 2: 63%

**Your mark:** \_\_\_\_\_ / 2

b. Equation of the circle

☐ Centre  $\frac{5}{2} + i$  (midpoint)

☐  $|z - (\frac{5}{2} + i)| = \frac{\sqrt{13}}{2}$

**Answer / notes:** Radius is half the diameter: some forgot to halve.

State average 1.2 / 2 (60%); 44% scored full marks. Mark distribution: 0: 30% · 1: 26% · 2: 44%

**Your mark:** \_\_\_\_\_ / 2

c. Sketch the circle

☐ Circle centre (1, 2), radius 2

☐ Imaginary-axis intercepts  $2 \pm \sqrt{3}$  labelled

**Answer / notes:** Draw it smoothly through the four extreme points.

State average 1.7 / 2 (85%); 74% scored full marks. Mark distribution: 0: 9% · 1: 17% · 2: 74%

**Your mark:** \_\_\_\_\_ / 2

d.i. Sketch the ray

- ☐ Ray from  $2 - i$  (open circle) through  $-2 + 3i$

**Answer / notes:** Start at  $2 - i$  with an open circle and extend through  $-2 + 3i$  and beyond.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

**Your mark:** \_\_\_\_\_ / 1

d.ii. Equation of the ray

- ☐  $\text{Arg}(z - (2 - i)) = \frac{3\pi}{4}$

**Answer / notes:** Direction  $-4 + 4i$  has argument  $\frac{3\pi}{4}$ .

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

**Your mark:** \_\_\_\_\_ / 1

e. Area of the minor segment

- ☐ Distance from centre to the line is  $\sqrt{2}$ , so the angle at the centre is  $\frac{\pi}{2}$
- ☐  $\frac{2^2}{2} \left( \frac{\pi}{2} - 1 \right) = \pi - 2$

**Answer / notes:**  $\pi - 2$ . Use the formula-sheet segment area with the right angle.

State average 1.3 / 2 (65%); 52% scored full marks. Mark distribution: 0: 27% · 1: 21% · 2: 52%

**Your mark:** \_\_\_\_\_ / 2

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**Question 4** (10 marks) · state average 7.3 / 10 (73%)

2025 VCAA Exam 2 · Section B · Question 1

a. Sketch with asymptotes, turning point and inflection

- ☐ Asymptotes  $x = -1$  and  $y = 0$  labelled
- ☐ Maximum  $(1, \frac{3}{4})$  labelled exactly
- ☐ Inflection point  $\approx (1.7, 0.6)$  labelled

**Answer / notes:** Denominator =  $(x + 1)(x^2 - x + 2)$ . Give the turning point exactly.

State average 2.1 / 3 (70%); 40.9% scored full marks. Mark distribution: 0: 6.97% · 1: 17.07% · 2: 35.07% · 3: 40.9%

**Your mark:** \_\_\_\_\_ / 3

b.i. Integral for the volume

- ☐  $\pi \int_0^2 \left( \frac{3x}{x^3 + x + 2} \right)^2 dx$

**Answer / notes:**  $V = \pi \int_0^2 y^2 dx$  with the function written in.

State average 0.9 / 1 (90%); 90.1% scored full marks. Mark distribution: 0: 9.9% · 1: 90.1%

**Your mark:** \_\_\_\_\_ / 1

b.ii. Volume to 2 d.p.

- ☐ 2.29

**Answer / notes:**  $\approx 2.29$  cubic units.

State average 0.8 / 1 (81%); 81.49% scored full marks. Mark distribution: 0: 18.51% · 1: 81.49%

**Your mark:** \_\_\_\_\_ / 1

c. Vertical asymptotes of  $\frac{3x}{x^3 - 5x + 2}$

- ☐  $x = 2, x = -1 + \sqrt{2}, x = -1 - \sqrt{2}$

**Answer / notes:**  $(x - 2)(x^2 + 2x - 1)$ . Write each asymptote as a separate equation.

State average 0.7 / 1 (73%); 73.53% scored full marks. Mark distribution: 0: 26.47% · 1: 73.53%

**Your mark:** \_\_\_\_\_ / 1

d.i. y-coordinate of the stationary point P

- ☐  $\frac{3}{a+3}$

**Answer / notes:** The derivative's numerator is  $6(1 - x^3)$ , so P is always at  $x = 1$ .

State average 0.8 / 1 (84%); 84.32% scored full marks. Mark distribution: 0: 15.68% · 1: 84.32%

**Your mark:** \_\_\_\_\_ / 1

d.ii. Vertical asymptotes when there are no stationary points

☐  $a = -3$ :  $x = 1$  and  $x = -2$

**Answer / notes:** No stationary point means  $x = 1$  is an asymptote:  $x^3 - 3x + 2 = (x - 1)^2(x + 2)$ .

State average 0.5 / 1 (45%); 45.85% scored full marks. Mark distribution: 0: 54.15% · 1: 45.85%

**Your mark:** \_\_\_\_\_ / 1

d.iii. a for an inflection at  $x = 2$

☐ Sets  $y''(2) = 0$

☐  $a = \frac{24}{5}$

**Answer / notes:** Show the second-derivative equation for both marks.

State average 1.5 / 2 (75%); 70.78% scored full marks. Mark distribution: 0: 20.12% · 1: 9.09% · 2: 70.78%

**Your mark:** \_\_\_\_\_ / 2

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**Question 5** (10 marks) · state average 7.4 / 10 (74%)

2025 VCAA Exam 2 · Section B · Question 2

a. Sketch  $z\bar{z} = 4$

☐ Circle centre O, radius 2

**Answer / notes:**  $|z|^2 = 4$ .

State average 0.9 / 1 (88%); 88.31% scored full marks. Mark distribution: 0: 11.69% · 1: 88.31%

**Your mark:** \_\_\_\_\_ / 1

b.i. Show the locus is  $y = \sqrt{3}x$

☐ Expands both moduli with  $z = x + iy$

☐ Simplifies to  $y = \sqrt{3}x$

**Answer / notes:** Full algebra (or a geometric midpoint-and-gradient argument) is needed.

State average 1.6 / 2 (80%); 74.45% scored full marks. Mark distribution: 0: 13.34% · 1: 12.21% · 2: 74.45%

**Your mark:** \_\_\_\_\_ / 2

b.ii. Sketch the line

☐ Line through O at  $60^\circ$  to the real axis

**Answer / notes:** The full line  $y = \sqrt{3}x$ , angled at exactly  $60^\circ$ .

State average 0.8 / 1 (78%); 78.3% scored full marks. Mark distribution: 1: 21.7%

**Your mark:** \_\_\_\_\_ / 1

c.i. Intersection points

☐  $4x^2 = 4$

☐  $1 + \sqrt{3}i$  and  $-1 - \sqrt{3}i$

**Answer / notes:** Substitute  $y = \sqrt{3}x$  into  $x^2 + y^2 = 4$ .

State average 1.5 / 2 (76%); 70.19% scored full marks. Mark distribution: 0: 18.93% · 1: 10.88% · 2: 70.19%

**Your mark:** \_\_\_\_\_ / 2

c.ii. Label the points

☐ Both points labelled

**Answer / notes:** Label as complex numbers or coordinates (no i inside coordinates).

State average 0.7 / 1 (73%); 73.89% scored full marks. Mark distribution: 0: 26.11% · 1: 73.89%

**Your mark:** \_\_\_\_\_ / 1

d.  $z_0$  and  $\theta$  for the ray from P through Q

☐  $z_0 = \frac{3}{2} + \frac{3\sqrt{3}}{2}i, \theta = -\frac{\pi}{3}$

**Answer / notes:** Use the principal argument.

State average 0.5 / 1 (50%); 50.64% scored full marks. Mark distribution: 0: 49.36% · 1: 50.64%

**Your mark:** \_\_\_\_\_ / 1

e. Area of the minor segment

☐ Angle at centre  $\frac{\pi}{3}$

☐  $\frac{9}{2} \left( \frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) = \frac{3\pi}{2} - \frac{9\sqrt{3}}{4}$

**Answer / notes:**  $c = \frac{3}{2}$ ,  $d = -\frac{9\sqrt{3}}{4}$ .

State average 1.4 / 2 (68%); 61.07% scored full marks. Mark distribution: 0: 23.49% · 1: 15.44% · 2: 61.07%

**Your mark:** \_\_\_\_\_ / 2

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**Question 6** (10 marks) · state average 7.5 / 10 (75%)

2025 VCAA Exam 2 · Section B · Question 6

a.i. Mean and sd of the sample mean

☐ Mean 1000, sd 16

**Answer / notes:** sd, not variance.

State average 0.9 / 1 (86%); 86.18% scored full marks. Mark distribution: 0: 13.82% · 1: 86.18%

**Your mark:** \_\_\_\_\_ / 1

a.ii. Probability the mean exceeds 970

☐ 0.9696

**Answer / notes:**  $\Pr(\bar{X} > 970)$  with  $\bar{X} \sim N(1000, 16^2)$ .

State average 0.8 / 1 (85%); 85.83% scored full marks. Mark distribution: 0: 14.17% · 1: 85.83%

**Your mark:** \_\_\_\_\_ / 1

b. 95% confidence interval

☐ (748.2, 751.8)

**Answer / notes:**  $750 \pm 1.96 \times \frac{5}{\sqrt{30}}$ .

State average 0.9 / 1 (88%); 88.09% scored full marks. Mark distribution: 0: 11.91% · 1: 88.09%

**Your mark:** \_\_\_\_\_ / 1

c. How many of 300 intervals contain  $\mu$

☐ 285

**Answer / notes:**  $0.95 \times 300$ .

State average 0.8 / 1 (76%); 76.91% scored full marks. Mark distribution: 0: 23.09% · 1: 76.91%

**Your mark:** \_\_\_\_\_ / 1

d. Minimum sample size

☐ 97

**Answer / notes:**  $n \geq 96.04$ : round up.

State average 0.5 / 1 (53%). Mark distribution: 0: 46.89%

**Your mark:** \_\_\_\_\_ / 1

e. Hypotheses

☐  $H_0 : \mu = 750$ ,  $H_1 : \mu < 750$

**Answer / notes:** The claim is “less than”, so one-tailed.

State average 0.9 / 1 (90%); 90.5% scored full marks. Mark distribution: 0: 9.5% · 1: 90.5%

**Your mark:** \_\_\_\_\_ / 1

f.i. p-value

☐ 0.0023

**Answer / notes:**  $\Pr(\bar{X} < 748)$  with  $\text{sd} = \frac{5}{\sqrt{50}}$ .

State average 0.8 / 1 (85%); 85.23% scored full marks. Mark distribution: 0: 14.77% · 1: 85.23%

**Your mark:** \_\_\_\_\_ / 1

**f.ii.** Is the company's claim correct?

☐ Yes:  $0.0023 < 0.01$

**Answer / notes:** Refer to the claim and the significance level.

State average 0.7 / 1 (68%); 68.04% scored full marks. Mark distribution: 0: 31.96% · 1: 68.04%

**Your mark:** \_\_\_\_\_ / 1

**g.** Critical sample mean

☐ 748.355

**Answer / notes:** Use the lower tail.

State average 0.6 / 1 (62%); 62.95% scored full marks. Mark distribution: 0: 37.05% · 1: 62.95%

**Your mark:** \_\_\_\_\_ / 1

**h.** Probability of a type II error

☐ 0.113

**Answer / notes:**  $\Pr(\bar{X} \geq 748.355 \mid \mu = 747.5)$ .

State average 0.5 / 1 (53%); 53.65% scored full marks. Mark distribution: 0: 46.35% · 1: 53.65%

**Your mark:** \_\_\_\_\_ / 1

## Your result

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**Total:** \_\_\_\_\_ / 118    State average on these questions: **85 / 118 (72%)**

**Topics to revise:**

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